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Realizing semicrystalline polymer physics across disparate processing scales: From desktop extrusion to large-format additive manufacturing

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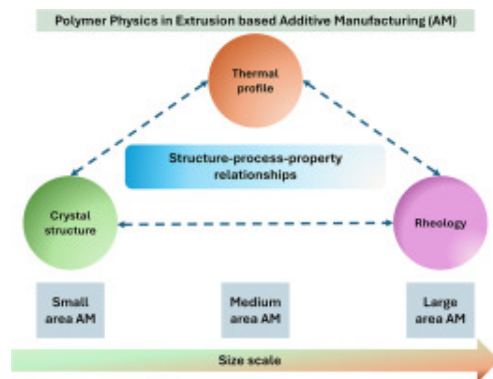
Abstract

Semicrystalline polymers pose unique advantages and challenges in translating from small-format extrusion-based additive manufacturing (EB-AM) to large-format EB-AM due to their complex crystallization behavior and the spatio-temporal thermo-mechanical property variations they exhibit during printing and solidification. In this work, we aim to synthesize the current state-of-the-art regarding semicrystalline polymers in EB-AM, with a focus on how polymer physics control process-structure-property relationships at disparate length- and time scales. The impact of crystallization kinetics, polymer chain mobility, and entanglement density on the evolution of printed

Loading [MathJax]/jax/output/SVG/jax.js interlayer diffusion is explored, thereby tying the link among molecular scale phenomena, mesoscale morphology development, and

macroscopic printed part properties. The widely different thermal and shear histories encountered during small- and large-format EB-AM can have a profound effect on the chain stretch and orientation, crystallization behavior, rheological response, and residual stress state during printing. Strategies to mitigate the exacerbated volumetric shrinkage and warpage issues for semicrystalline polymers in EB-AM through material design and processing modifications are highlighted. Such approaches are critical to not only ensure dimensionally accurate parts but also minimize anisotropy and poor interlayer adhesion. Routes to monitor spherulitic growth and melt-viscosity variations during EB-AM, such as *in-situ* scattering techniques and infrared thermography, provide insights needed to achieve consistent and reliable processing. Finally, we identify the fundamental research gaps that currently plague semicrystalline polymers in EB-AM and highlight the importance of developing predictive multiscale modeling frameworks coupled with formulation of next-generation material systems and process monitoring capabilities to enable widespread adoption of the technique.

Graphical abstract



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Large-format additive manufacturing; Fused filament fabrication; Semicrystalline polymers; Thermal history; Crystallization kinetics; Interlayer adhesion

Abbreviations

ABS, Acrylonitrile butadiene styrene; AM, Additive manufacturing; ASTM, American Society for Testing and Materials; $\theta(t)$, Angle of intersection; $\dot{\gamma}_{aw}$, Apparent shear rate at the nozzle wall; U_{ex} , Average extrusion speed; $\dot{\gamma}$, Average shear rate in the nozzle; BAAM, Big area additive manufacturing; CF, Carbon fiber; CNT, Carbon nanotubes; ChNCs, Chitin nanocrystals; CTE, Coefficient of thermal expansion; η^* , Complex viscosity; CCF, Continuous carbon fiber; $t_{0.5}^{crys}$, Crystallization half-time; De, Deborah number; T_d , Degradation temperature; τ_{des} , Deposition time; DIW, Direct ink write; ε , Extensional strain; η_e , Extensional viscosity; EB-AM, Extrusion based additive manufacturing; v , Feed velocity; FEA, Finite element analysis; σ_1 , First normal stress difference; ω , Frequency; FFF, Fused filament fabrication; τ_ϕ , Gelation time scale; GB, Glass beads; GF, Glass fiber; T_g , Glass transition temperature; GNP, Graphene nanoplatelets; $H(t)$, height of the deposited road; HDPE, High density polyethylene; HA, Hydroxyapatite; ρ_{voids} , Inter-track void density; iPP, Isotactic polypropylene; LAAM, Large area additive manufacturing; LSAM, Large scale additive manufacturing; t_L , Layer time; L_{fil} , Length of filament deposited on print bed; L_{noz} , Length of nozzle; G'' , Loss modulus; MAAM, Medium area additive manufacturing; MCC, Microcrystalline cellulose; MWCNT, Multiwalled carbon nanotube; R , Nozzle radius; $T_{d,onset}$, Onset of degradation temperature; T_c , Peak crystallization temperature; T_m , Peak melting temperature; PBS, Polybutylene succinate; PCL, Polycaprolactone; PC, Polycarbonate; PEEK, Polyether ether ketone; PEI, Polyether imide; PEKK, Polyether ketone ketone; PES, Polyethersulfone; PET, Polyethylene terephthalate; PETG, Polyethylene terephthalate glycol; PHA, Polyhydroxyalkanoate; PLA, Polylactic acid; PPS, Polyphenylene sulfide; PPSU, Polyphenylsulfone; PBF, Powder bed fusion; n , Power law index; τ_d , Reptation time; Wi_d , Reptation Weissenberg number; τ_{res} , Residence time in the nozzle; Re , Reynolds number; τ_R , Rouse time; Wi_R , Rouse Weissenberg number; s , Loading [MathJax]/jax/output/SVG/jax.js shear stress; SAXS, Small angle X-ray scattering; SAAM,

Small area additive manufacturing; η , Steady shear viscosity; G' , Storage modulus; $\bar{\gamma}$, Strain amplitude; T , Talcum; α , Thermal diffusivity; ∇T , Thermal gradient; TPU, Thermoplastic polyurethane; 3D, Three dimensional; t , Time; τ_α , Time scale for achieving uniform temperature in the nozzle; τ_c , Time taken by the polymer melt to reach the crystallization temperature; τ_g , Time taken by the polymer melt to reach the glass transition temperature; U_{noz} , Transverse nozzle speed; VP, Vat photopolymerization; Q , Volumetric flow rate; $\dot{\gamma}_w$, Wall shear rate; WAXS, Wide angle X-ray scattering; $w(t)$, Width of the road; WF, Wood flour; ZrO_2 , Zirconium oxide

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Data availability

No data was used for the research described in the article.

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